

Ensembl data resources for variant interpretation Workshop



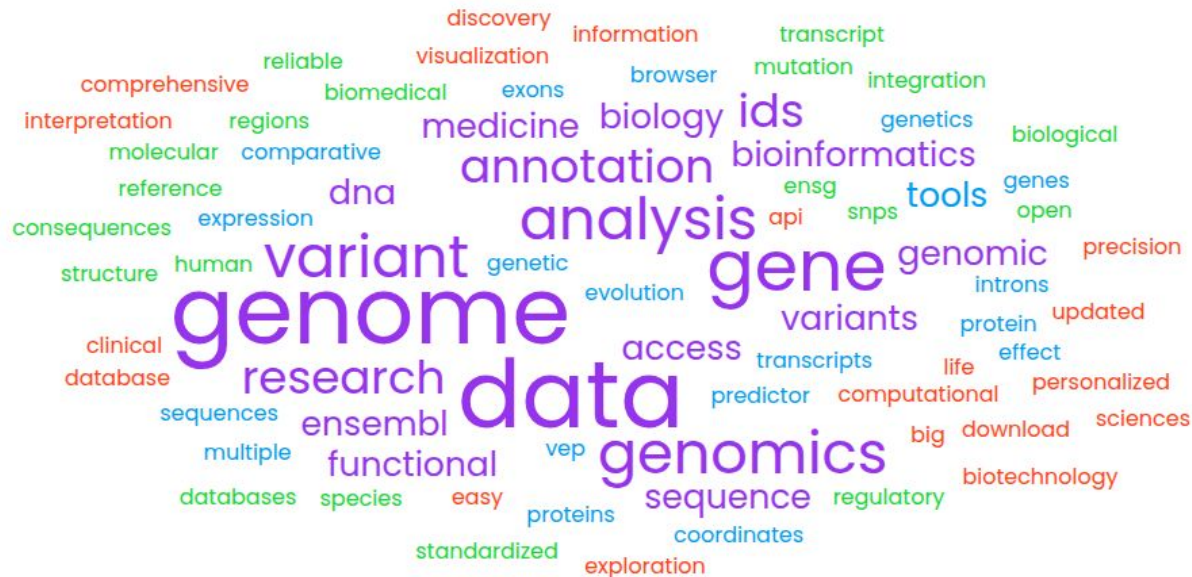
Clinical Bioinformatics for Variant Interpretation in Medical Genomics

International Centre for Genetic Engineering and Biotechnology

Firas Zemzem

**Higher Institute Of Biotechnology Of Monastir -
Tunisia**

06th October 2026

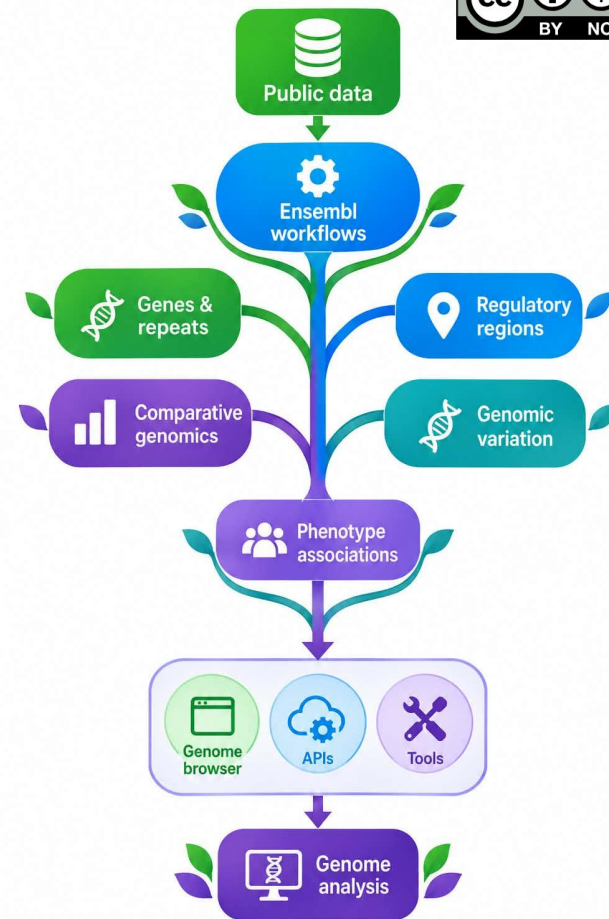


source

How Ensembl enables genome analyses !!



- > Study genes and repetitive sequences to identify and characterize genomic elements.
- > Regulatory regions can be investigated to understand how gene expression is controlled.
- > Identify conserved or evolutionarily related regions.
- > Differences between individuals or populations. Finally, phenotype associations connect genetic information with observable traits or biological characteristics.

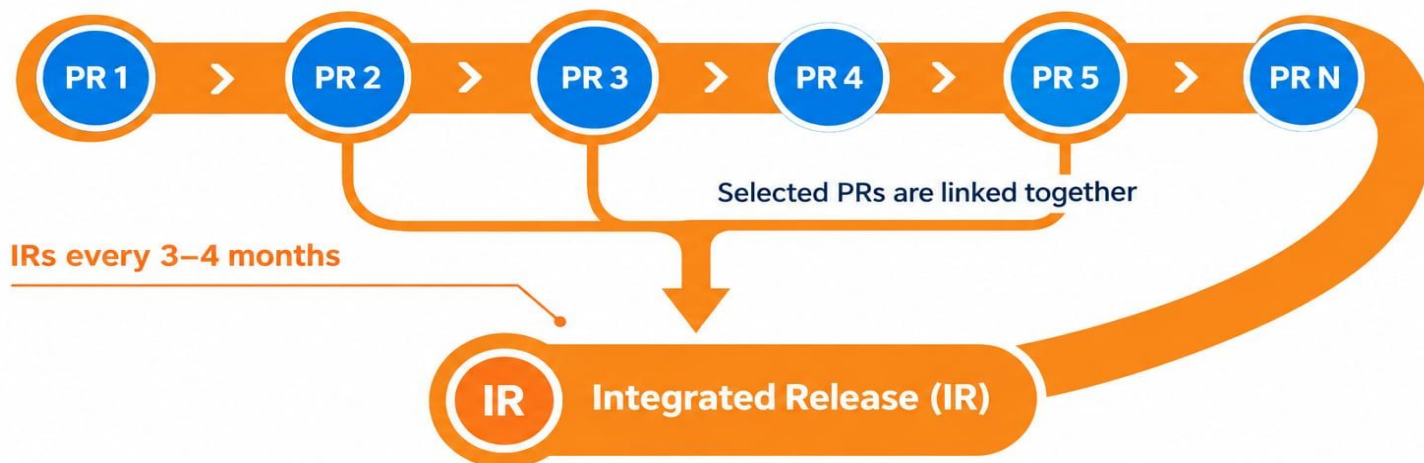


Partial Releases (PRs)

For rapid access: new assemblies, gene sets, variation and regulation data released every 2 weeks.

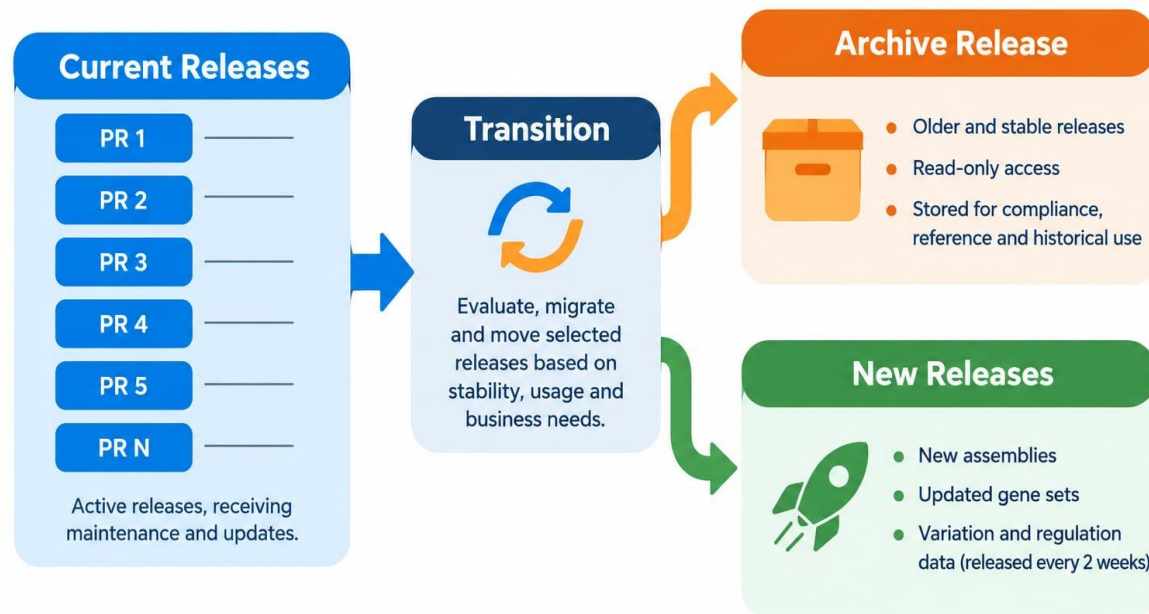
Integrated Releases (IRs)

For long-term stability: frozen snapshots of selected PRs suitable for reproducible research and citations.



> Stable reference point <





Ensembl is a public and open project providing access to genomes, annotations, tools and methods. Its goal is to enable genomic science by providing high-quality, integrated and consistent annotation on all cellular genomes within a harmonious, scalable and accessible infrastructure.

Ensembl Release 116 (June 2026)

- Ensembl 116 is the final release on this platform. All new data will be available on the new Ensembl site.
- New flag - "Ensembl Canonical Extended" applied to human transcripts with extended 5' and 3' ends.
- New pig breeds, cattle, and donkey genomes have been added.
- The EPO-Extended comparative alignment now includes 30 pigs.
- Duck, dromedary camel and Atlantic herring now have variation support.
- Regulation data for Atlantic salmon has been updated.
- All external references have been updated and InterProScan has been rerun on all species across Ensembl divisions.

[More release news](#) on our blog

The New Ensembl Site

The new Ensembl provides access to over 4700 genomes: <http://beta.ensembl.org>

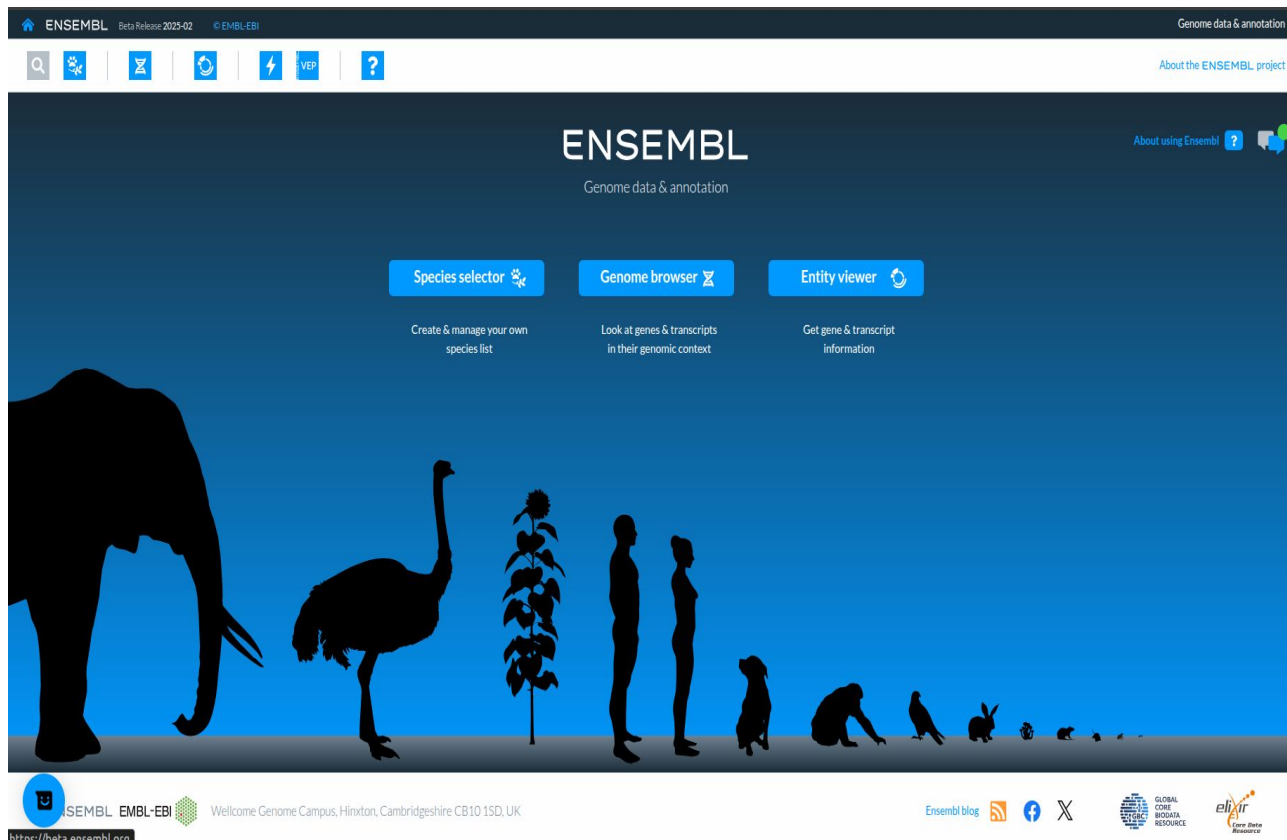
There are over 4100 animal, 470 plant, and 100 fungal genomes ready to explore. We provide pangenomes for many species including human (565 haplotypes), barley (69 cultivars), and pig (27 breeds).

Genomes from projects such as Darwin Tree of Life, the Human Pangenome Reference Consortium, the Vertebrate Genomes Project and the European Reference Genome Atlas are regularly added.

We are in the process of migrating our existing prokaryotic genomes, with a target of moving all ~36,000 by July 2026. Prokaryotic releases will be phased, with batches migrating from Feb 2026 onwards.

From the summer of 2026, all new data will only be available through this new Ensembl site. Important info for programmatic access is available on this [blog](#).





The screenshot shows the Ensembl website interface. At the top, there is a dark blue header with the Ensembl logo, "Beta Release 2025-02", and "© EMBL-EBI". On the right, it says "Genome data & annotation". Below the header is a navigation bar with icons for search, species selector, genome browser, entity viewer, and a help icon. The main content area has a dark blue background with the Ensembl logo and "Genome data & annotation". Below this are three main buttons: "Species selector" (with a paw print icon), "Genome browser" (with a magnifying glass icon), and "Entity viewer" (with a circular arrow icon). Each button has a description below it: "Create & manage your own species list", "Look at genes & transcripts in their genomic context", and "Get gene & transcript information". At the bottom, there is a large illustration of various organisms in silhouette, including an elephant, an ostrich, a sunflower, two human figures, a dog, a monkey, a bird, a rabbit, and several small animals. The footer contains the Ensembl logo, "EMBL-EBI", "Wellcome Genome Campus, Hinxton, Cambridgeshire CB10 1SD, UK", and a URL "https://beta.ensembl.org". On the right, there are links to the Ensembl blog, social media icons for YouTube, Facebook, and X, and logos for the Global Core Biodata Resource and Elixir.



Site search

Genome selector

Genome Browser

Alignments viewer

Feature explorer

Blast

Ensembl VEP

ENSEMBL

Release 2026-07

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Genome data & annotation

Archives

Contact us

About the ENSEMBL project

Genome selector

Genomes in use

1 / 1

Manage

Human

GRCh38.p14

Select a tab to see a Genome home page

Find a genome

Common or scientific name...

Find

Find a feature in the selected genomes

Gene, transcript or variant ID...

Find

8,236 genomes now available

4,593 Animals

110 Archaea

2,423 Bacteria

388 Fungi

573 Green Plants

149 Others

Popular species

External projects

species Consortia - projects

563

30

4

22

20

2

6

5

30

5

9

69

18

22

2

2

4

2

3

2

3

4

3

3

2

3

4

3

53

18

Popular species

Release 2026-07 - Firas Zemzem - 06th October 2026 - Tunisia - ICGEB International Course



About the ENSEMBL project

Genome selector

Genomes in use 1 / 1 [Manage](#)

Human GRCh38.p14

Add a genome

Close

Help



Human GRCh38.p14 **563** Assemblies

Don't use Use

Remove from list

Coding genes	Coding genes	Shortest coding gene	Longest coding gene	Example gene
	20,543	8 bp	2,473,539 bp	
	Transcripts in coding genes	Exons in coding genes	Introns in coding genes	
	430,914	4,334,504	3,903,590	
Analysis	Average coding genomic span	Average coding sequence length	Average CDS length	
	69,231.86 bp	3,562.81 bp	1,234.08 bp	
	Average transcripts per coding gene	Average exon length per coding gene	Average intron length per coding gene	
	20.98	285.14 bp	5,431.5 bp	
Transcripts	Coding transcripts	Average coding transcripts per gene		
	370,915	18.06		
	Coding exons	Average coding exon length		
	3,215,032	142.37 bp		
Non coding genes	42,155			

[Download](#)

Assembly

DNA sequence

FASTA

Annotation

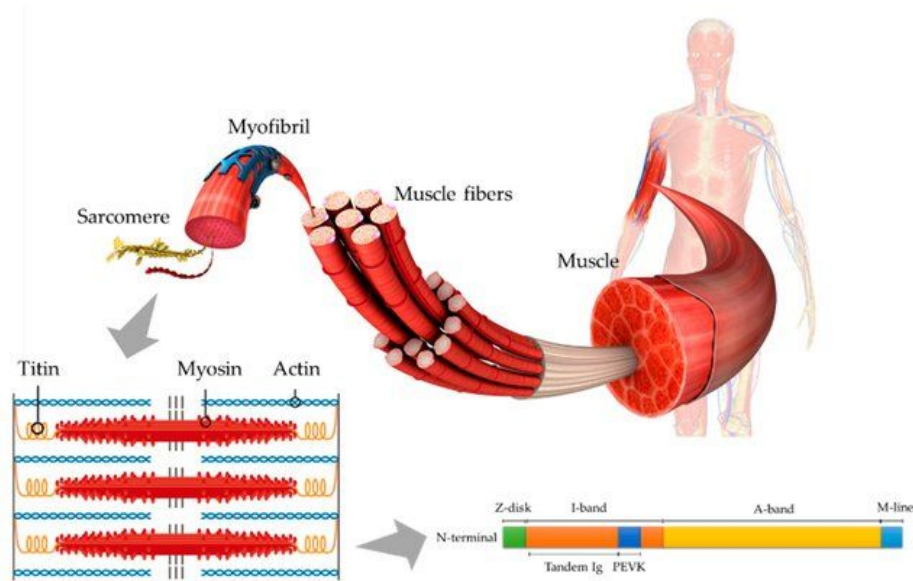
Genes, cDNA, ncRNA, proteins

FASTA/GTF/GFF3

Homologies

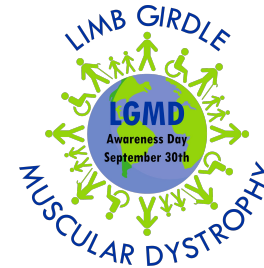
TSV

Case study - limb-girdle muscular dystrophy type 2A (LGMD2A)



(Image source: 123RF)

This is a schematic illustration of titin in the muscle. Muscle comprises of muscle fibers, myofibrils, and the smallest units, known as sarcomeres. In the sarcomere (lower left), titin connects actin-containing thin filaments and myosin-containing thick filaments. ([Nakanishi N et al.](#))



> Calpain 3, encoded by the **CAPN3** gene, is a muscle-specific protease that binds to titin and plays a key role in muscle function.

> Mutations in **CAPN3** cause **LGMD2A**, characterized by progressive weakness of proximal muscles.

> Diagnosis combines genetic testing with protein assays, as LGMD2A likelihood correlates with calpain 3 deficiency.



LGMD



Case study - limb-girdle muscular dystrophy type 2A (LGMD2A)



ENSEMBL Beta Release 2025-02 © EMBL-EBI

Species Selector Species in use 1 / 1 Manage

Human GRCh38.p14 Close

Find a gene

CAPN3

Go

☒ Only species in list ☐ All Ensembl

Found in 1 species

	Gene
Human GRCh38.p14	CAPN3 ENSG00000092529.26

transcript-identifier_transcript-Version

(docs)



Case study - limb-girdle muscular dystrophy type 2A (LGMD2A)



ENSEMBL Release 2024-07 EMBL-EBI

Genome data & annotation Archives Contact us

About the ENSEMBL project

Genome browser Genomes in use (1 / 1) Manage

Human GRCh38.p14

Gene CAPN3 ENSG00000092529.27

bp 30,000 42,340,000 42,350,000 42,360,000 42,370,000 42,380,000 42,390,000 42,400,000 42,410,000 42,420,000 42,430,000 42,440,000

42,359,494 42,412,949

CAPN3 ENSG00000092529

G

15:42,332,767-42,439,676

Go

Genomic Variation Regulation

CAPN3 protein_coding
ENST00000397163.8 MANE Select
+59 transcripts

Find CAPN3 protein_coding
ENST00000397163.8 MANE Select
+59 transcripts

Find Short variant groups

Protein altering variants

Splicing variants

Transcript variants

Regulatory region variants

Intergenic variants

Short variants by resource

dbSNP short variants

Protein altering variants

SO term	SO accession
feature_elongation	SO:0001907
feature_truncation	SO:0001906
frameshift_variant	SO:0001589
inframe_deletion	SO:0001822
inframe_insertion	SO:0001821
missense_variant	SO:0001583
protein_altering_variant	SO:0001818
start_lost	SO:0002012
stop_gained	SO:0001587
stop_lost	SO:0001578
transcript_ablation	SO:0001893
transcript_amplification	SO:0001889

ENSEMBL Release 2024-07 EMBL-EBI

Genome browser Genomes in use (1 / 1) Manage

Human GRCh38.p14

Gene CAPN3 ENSG00000092529.27

CAPN3 protein_coding
ENST00000397163.8 MANE Select
+59 transcripts

Find CAPN3 protein_coding
ENST00000397163.8 MANE Select
+59 transcripts

Region 15

Go to Add location co-ordinates...

Assembly

Reference sequence

%GC

Transcription start sites

CpG islands

Repeats: Repeatmasker RefSeq human

Low complexity: Dust

Repeats: Centromere

Repeats: TRF

Sequences S3456 top

53456 top

15:42,359,494-42,412,949

Go

Blast whole sequence

Genomic sequence

Reverse complement

Download



Case study - limb-girdle muscular dystrophy type 2A (LGMD2A)



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Genome data & annotation Archives Contact us

About the ENSEMBL project

Feature explorer Genomes in use 1 / 1 Manage

Human GRCh38.p14

Gene CAPN3 ENSG00000092529.27 Biotype protein_coding forward strand 15:42,359,494-42,412,949

Overview External references

5' 3' forward strand 60 transcripts

bp 1 10,000 20,000 30,000 40,000 50,000

Filter & sort Transcripts Transcripts table Gene function Gene relationships

All transcripts

Transcript	cDNA length (bp)	Protein length (aa)	Biotype	CCDS	UniProt match	RefSeq match	Flags
ENST00000397163.8	3,315	821	protein_coding	CCDS45245	P20807	NM_000070.3	Ensembl canonical MANE Select GENCODE basic APPRIS P1 TSL-1
ENST00001132962.1	3,322	821	protein_coding	CCDS45245	—	—	GENCODE basic APPRIS P1
ENST00000674093.1	1,984	156	protein_coding	CCDS10086	P20807	—	GENCODE basic
ENST00000673890.1	1,416	156	protein_coding	CCDS10086	P20807	—	GENCODE basic
ENST00000337571.9	1,304	156	protein_coding	CCDS10086	P20807	—	GENCODE basic TSL-1
ENST00000673936.1	1,317	156	protein_coding	CCDS10086	P20807	—	GENCODE basic
ENST00000674052.1	2,354	229	protein_coding	—	—	—	GENCODE basic

Gene name: CAPN3 ENSG00000092529.27

calpain 3 HGNC:1480

Synonyms: CAPN3, LGMD2, LGMD2A, NCL-1, P94

Attributes: Biotype: protein_coding

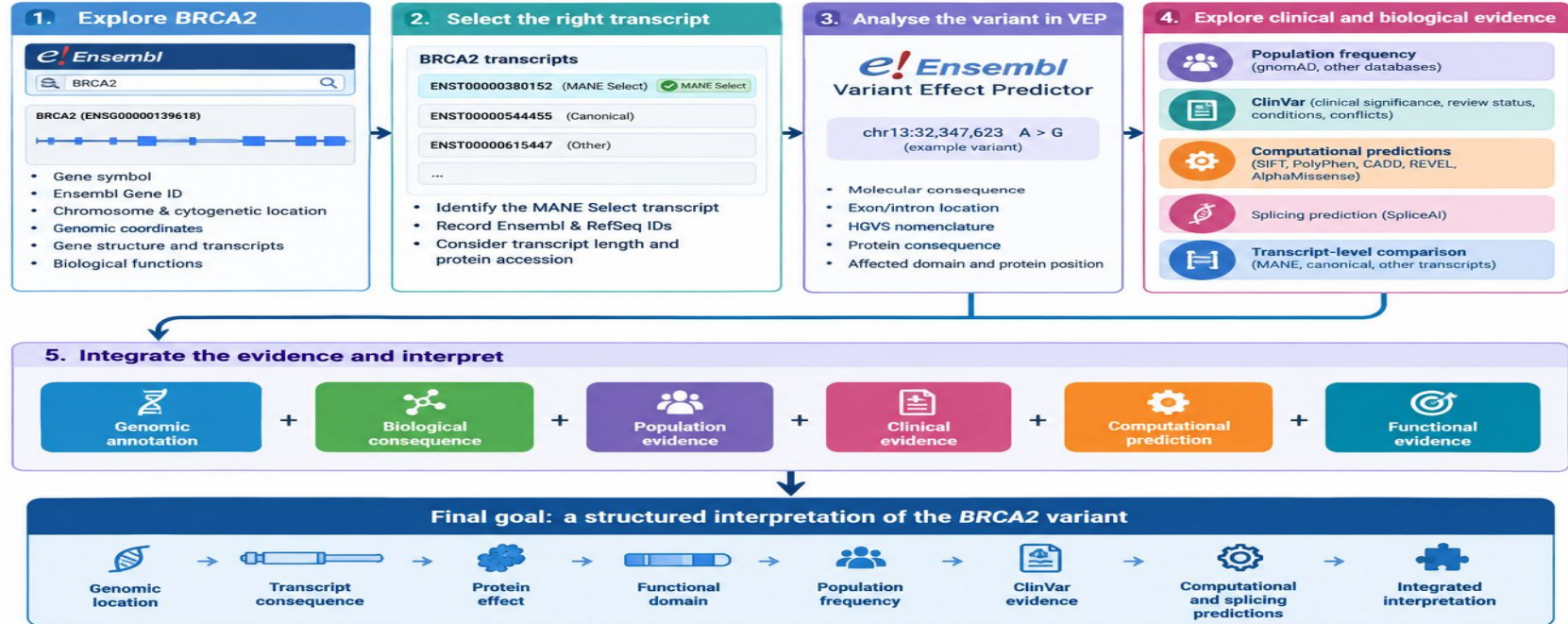
Find

Function: Other data sets

Publications: Europe PMC



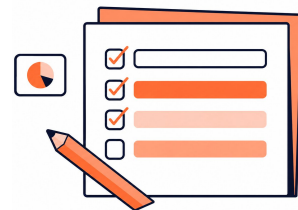
Case study - Find BRCA2 Gene



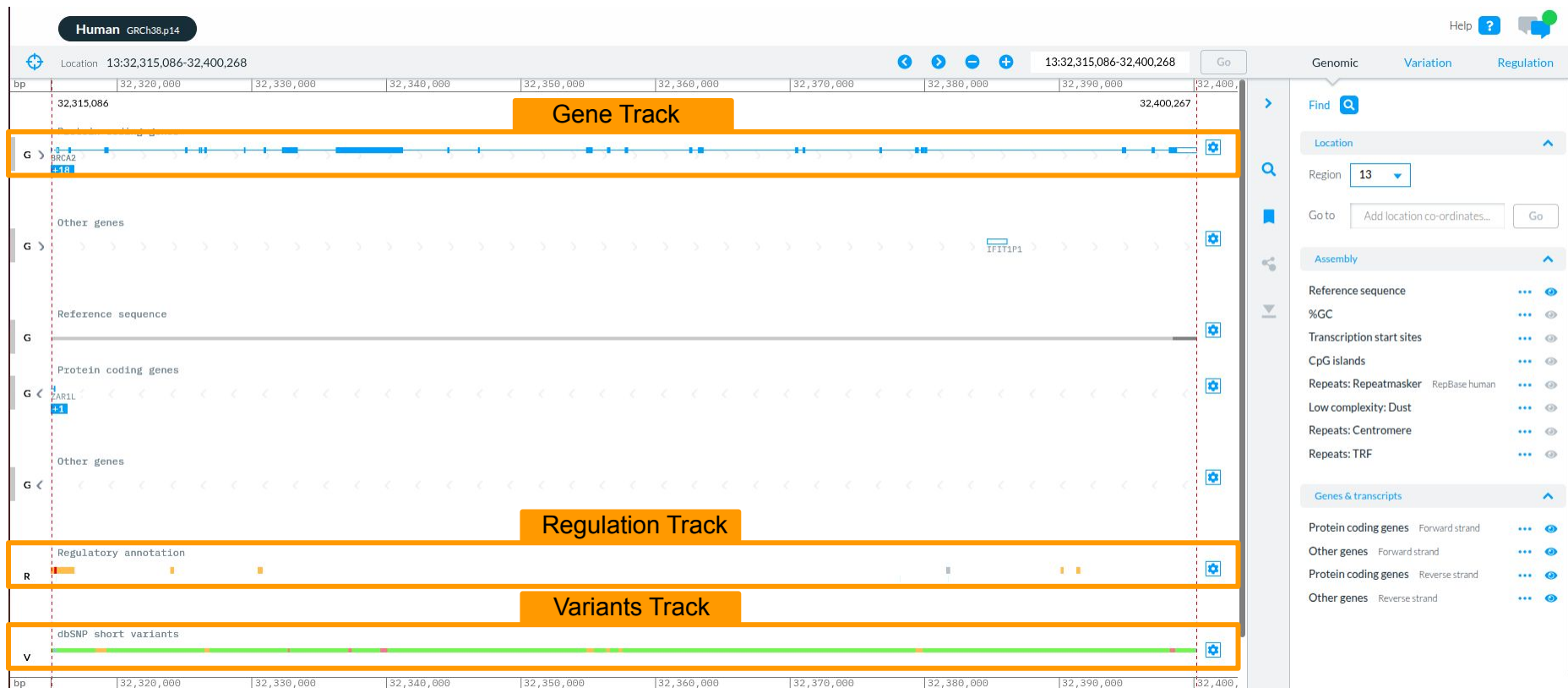
Questions



- * What is the Ensembl Gene ID for BRCA2?
- * On which chromosome is BRCA2 located?
- * What is its cytogenetic location?
- * What are its GRCh38 genomic coordinates?
- * What is the gene orientation?
- * What is the BRCA2 MANE Select transcript?
- * What is the corresponding RefSeq transcript?
- * What is the protein accession and protein length?
- * What are the main biological functions of BRCA2?
- * Why is the MANE Select transcript useful for variant interpretation?



Case study - Find BRCA2 Gene



Case study - Find BRCA2 Gene



- * **Gene:** BRCA2
- * **Ensembl ID:** ENSG00000139618
- * **Chromosome:** 13
- * **Cytogenetic location:** 13q13.1
- * **Genomic coordinates:** chr13:32,315,086-32,400,268

- > Use Go to box to switch location to the target gene
- > Use Genomic coordinates
- > **Hint :** Do not use chromosome name - just use chr number

The screenshot shows the Ensembl genome browser interface. The top navigation bar includes tabs for Genomic, Variation, and Regulation. The main content area displays the location of the gene on chromosome 13. The 'Region' dropdown is set to 15. The 'Go to' box contains the genomic coordinates .3:32,315,086-32,400,268, and a green 'Go' button is next to it. Below the 'Go to' box, the 'Assembly' section is expanded, showing various genomic features such as Reference sequence, %GC, Transcription start sites, CpG islands, Repeats: Repeatmasker, Low complexity: Dust, Repeats: Centromere, and Repeats: TRF. The 'Genes & transcripts' section is also expanded, showing Protein coding genes and Other genes for both Forward and Reverse strands.

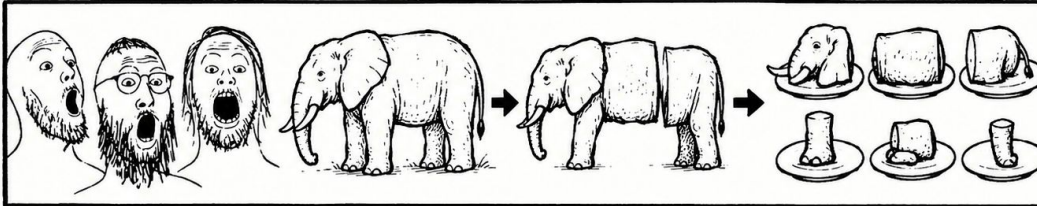
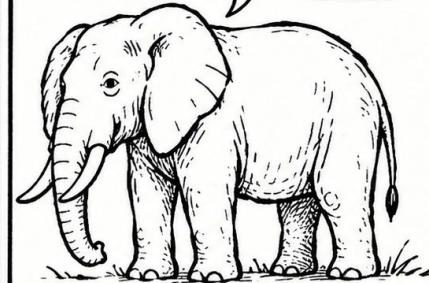


"YOU COULD NEVER EAT AN ELEPHANT
IN ONE TIME, YOU HAVE TO SPLIT IT
INTO PIECES."

YOU COULD NEVER EAT
AN ELEPHANT IN ONE TIME,



YOU HAVE TO SPLIT IT
INTO PIECES."



> In this practical live assignment, we will investigate the SHH (Sonic Hedgehog) gene using the new ensembl platform release. We will first identify and **characterize the SHH locus** and then **explore its transcripts** in detail to understand their structure and expression. We will also examine naturally occurring **genetic variation** in SHH, including differences observed across populations. Finally, we will investigate external **biological databases** to integrate and interpret additional information about the SHH gene and its functional significance.

Original Article

The role of the sonic hedgehog signalling pathway in patients with midline defects and congenital hypopituitarism

[L.C. Gregory](#), [C. Gaston-Massuet](#), [C.L. Andoniadou](#), [G. Carreno](#), [E.A. Webb](#), [D. Kelberman](#), [M.J. McCabe](#), [L. Panagiotakopoulos](#), [J.W. Saldanha](#), [H.A. Spoudeas](#), [J. Torpiano](#), [M. Rossi](#), [J. Raine](#), [N. Canham](#), [J.P. Martinez-Barbera](#), [M.T. Dattani](#) 

First published: 17 October 2014 | <https://doi.org/10.1111/cen.12637> |

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Please instruction answer here :

> https://drive.google.com/file/d/1NNMRoz7WdN72OzaO1d6_qC4lnyY750tN/view?usp=sharing



Thank you

If you have any questions or would like further information, please feel free to get in touch with me.
zemzemfiras@gmail.com



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